



SKILLCLAW FRAMEWORK



# SkillClaw

**Let Skills Evolve Collectively**  
with Agentic Evolver



Collective Skill Evolution for Multi-User LLM Agent Ecosystems



# Agent Skills Remain Static After Deployment — Users Rediscover Solutions Independently



## Static Skill Paradigm

- Skills manually installed from hub
- Never updated from runtime experience
- Same failures rediscovered repeatedly
- Memory-based methods tied to instances
- Skill libraries treated as static resources

No System-Level Learning

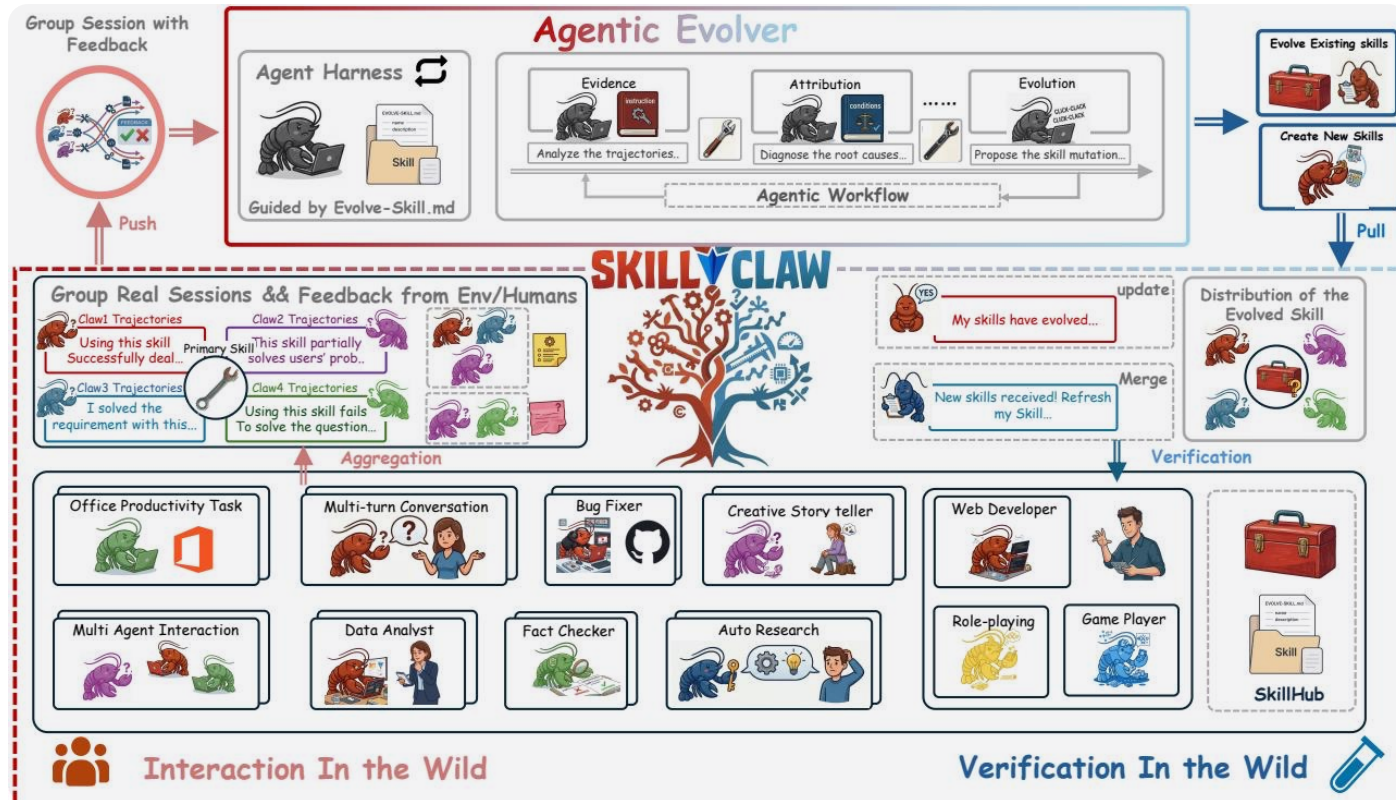


## Evolving Skill Paradigm

- Skills improve from interaction evidence
- Cross-user knowledge accumulation
- Solutions persist beyond sessions
- Shared behavioral boundary discovery
- Continuous, automatic improvement

Collective Skill Evolution

# Closed-Loop Pipeline: From Multi-User Interaction to Shared Skill Evolution



## AGENTIC EVOLVER PIPELINE



### 1. Evidence

Analyze trajectories for behavioral patterns



### 2. Attribution

Diagnose root causes of failures & successes



### 3. Evolution

Propose skill mutations — refine or create



### 4. Distribution

Verify, merge & synchronize to all agents

Multi-User Interaction → Session Collection → Skill Evolution → Skill Synchronization

# Agentic Evolver: Evidence → Attribution → Evolution in Three Stages



## 1. Evidence

### Behavioral Pattern Analysis

Analyze trajectories to identify behavioral patterns across user sessions.

Structured action–feedback causal chains preserved.

Pattern Mining



## 2. Attribution

### Root Cause Diagnosis

Diagnose root causes of failures and successes from aggregated evidence.

Cross-user context reveals boundary conditions.

Causal Analysis



## 3. Evolution

### Skill Mutation Proposal

Propose skill mutations — refining existing skills or creating new ones.

Open-ended reasoning, NOT predefined rules.

Autonomous Agent

The evolution process is driven by an autonomous agent performing open-ended reasoning over interaction evidence

# Cross-User Trajectory Aggregation Exposes Skill Behavioral Boundaries



## Compatible Claw-Style Systems

OpenClaw · CoPaw · IronClaw

PicoClaw · ZeroClaw · NanoClaw

NemoClaw

### How it works:

- Independent agents produce structured session trajectories
- Trajectories aggregated by referenced skill
- Full action–feedback causal chains preserved



## Why Cross-User Aggregation Matters

Different users exercising the same skill under diverse contexts produce **complementary views** of that skill's behavioral boundary.

### Key Insight:

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

### Reveals both:

✓ Conditions under which the skill works

✗ Conditions under which the skill breaks

# Multimodal Creative Pipeline Skill

## — 6 Days of Evolution Attempts

| Day | Candidate Skill                                   | Skill Function                                    | Validator | Next-Day Action            |
|-----|---------------------------------------------------|---------------------------------------------------|-----------|----------------------------|
| 1   | validate-tmp-workspace-inputs                     | Check /tmp_workspace inputs & env setup           | Accept    | Upgrade to new best pool   |
| 2   | multimodal-input-validation-and-setup             | Multimodal input validation & output env init     | Reject    | Keep current best pool     |
| 3   | multimodal-creative-task-pipeline                 | Multimodal creative pipeline (PDF/video/image)    | Reject    | Keep current best pool     |
| 4   | multimodal-creative-task-pipeline (impr.)         | Added image classification, visual generation     | Reject    | Keep current best pool     |
| 5   | creative-pipeline + validate-required-input-files | Creative pipeline + per-file fail-fast validation | Reject    | Keep current best pool     |
| 6   | multimodal-creative-task-pipeline (cand.)         | Extended PDF-to-poster / document-to-visual       | Reject    | Not admitted to next cycle |

**Result:** Only 1 of 6 candidates accepted

The conservative verification process prevents regressions. Most proposed mutations failed to demonstrate measurable improvement over the existing best pool, so the system retained the proven skill.

# Git Push Auth-Fallback Skill

## — Iterative Refinement Over 6 Days

| Day | Candidate Skill                   | Change Summary                                               | Validator | Next-Day Action            |
|-----|-----------------------------------|--------------------------------------------------------------|-----------|----------------------------|
| 1   | git-push-with-auth-fallback       | Safe fallback instead of blocking on push failure            | Accept    | Add to Safety best pool    |
| 2   | git-push-with-auth-fallback       | Unified patch/bundle filenames, reduced inconsistency        | Accept    | Keep updated best pool     |
| 3   | git-push + git-clone-to-directory | Auth-alternative paths & secrets audit; fixed subshell       | Accept    | Keep current best pool     |
| 4   | (none)                            | Same-pool retest; validator confirmed current pool best      | Accept    | Continue same best pool    |
| 5   | git-push-with-auth-fallback       | Added "push hang treated as auth failure"; no improvement    | Reject    | Keep current best pool     |
| 6   | git-push-with-auth-fallback       | Added identity config & filename consistency; no improvement | Reject    | Not admitted to next cycle |

**Result:** 3 of 6 days yielded accepted updates

Successful iterative refinement: the skill was improved three times before evidence showed the best pool had plateaued. Days 5–6 produced no measurable gain, so the system halted further mutation attempts.

# Three Properties That Distinguish SkillClaw from Existing Systems



## Collective Evolution

Shared, Continuously Improving Ecosystem

Knowledge from individual interactions contributes to a shared skill ecosystem that improves continuously across users.

Cross-User

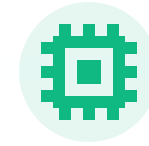


## Fully Automatic

No Manual Curation Required

Skill evolution is driven by runtime interaction without manual curation or explicit user intervention.

Zero-Touch



## Agentic Evolution Paradigm

Open-Ended Reasoning Over Evidence

Skill updates are produced through open-ended reasoning rather than predefined update rules.

Autonomous Agent

### Implications for Agent Systems:

Compatible with OpenClaw, CoPaw, IronClaw, PicoClaw, ZeroClaw, NanoClaw, NemoClaw.

Evaluation on WildClawBench (Qwen3-Max) demonstrates substantial improvements across tasks.